Summer Training Report

On

"INTERPRETATION OF DEPOSITIONAL ENVIRONMENT FROM LITHOFACIES AND WELL LOG ANALYSIS IN KG BASIN"

INTERNSHIP WORK DONE AT

KDMIPE, ONGC,

DEHRADUN

Submitted By:

RISHANT PRAKASH
BSc(H) GEOLOGY
Hansraj College
University of Delhi

Supervised by:

RAJESH SHARMA
GM(GEOLOGY)
I/c KG-PG BASIN, BRG-II

July, 2019

CERTIFICATE

Oil & Natural Gas Corporation Limited
Basin Research Group,
Keshava Deva Malviya Institute of Petroleum Exploration, Dehradun-248001

To Whom It May Concern

This is to certify that **Mr. Rishant Prakash** student of B.Sc. Geology (Hons), from Hansraj college, University of Delhi has completed his one-month summer training in the session 2019-20 from 11th June to 10th July under my supervision at KG-PG Basin Group, **KDMIPE, ONGC, DEHRADUN**. His training project work entitled as **"INTERPRETATION OF DEPOSITIONAL ENVIRONMENT FROM LITHOFACIES AND WELL LOG ANALYSIS IN KG BASIN"** embodies the result of the work carried out during the period of training. His work was carefully reviewed and corrected and he is entitled to submit his work for the successful completion of his summer training.

I wish him all the best for all his future endeavours.

Date: ……
Place:……..

RAJESH SHARMA
GM (Geology)
KDMIPE, ONGC,

ACKNOWLEDGEMENT

With deep sense of gratitude first of all I would like to thank my mentor **Mr. Rajesh Sharma,** GM (GEOLOGY), KG GROUP at **KDMIPE, ONGC Dehradun** for his support and guidance at each and every stage of my summer training. I am also indebted to **Ms Nishtha Agarwal** Geologist **& Mr. Soumen Deshmukh,** Sr. Geophysicist, KG GROUP); KDMIPE ONGC DEHRADUN for giving their continuous help and support for successfully completing my work.

I would like to express my sincere gratitude to **Jayanta Kumar Mishra,** General Manager (HR), KDMIPE Library who helped me select the correct choice of books and all the library mates who helped me keep my zeal throughout the training.

I am also in short of words expressing my heartiest thanks to my parents, my family members and close friends whose blessing and affection have been remained as a constant source of inspiration. Without the help of all these people, it would not have been possible for me to complete this work and submit this report on time.

CONTENT

PART A: INTRODUCTION

- **DEPOSITIONAL ENVIRONMENTS**.....2
  - **FLUVIAL SYSTEM**.....2-4
  - **DELTAIC SYSTEM**……...5-6
  - **BEACH & BARRIER ISLAND SYSTEM**……...7-8
  - **ESTUARINE SYSTEM**……..9
  - **LAGOONAL SYSTEM**……..10
  - **TIDAL FLAT SYSTEM**…….11
- **WELL LOGS**
  - **GAMMA RAY**…….12
  - **RESISTIVITY LOG**……..13

PART B: CASE STUDY

- **LOCATION & GEOLOGY OF STUDY AREA**……14
- **STRATIGRAPHY OF STUDY AREA**……...15-17
- **INTERPRETATION OF DEPOSITIONAL ENVIRONMENT AND LITHOLOGY OF STUDY AREA WELLS**.....18-26

PART C:

- **CONCLUSION**……27
- **REFERENCES**…….28

INTRODUCTION

This report presents the interpretation of depositional environment from lithofacies and well log analysis in Krishna-Godavari basin. In order to properly understand and carry out the work, pre-requisite knowledge of lithology, grain size, porosity, sorting, matrix content and sedimentary structures in fluvial systems, deltaic system, beach and barrier island system, estuarine system, lagoonal system and tidal-flat system have been collected from various books and internet. Also, basic knowledge on the interpretation of the lithology and depositional environment from gamma ray and resistivity log characteristics have been collected.

The study has been carried out in the synrift section of the offshore field of the KG basin. In this study, the determination of the lithology of the synrift sequence has been done on the basis of gamma ray characteristics. Depositional environment and age of the synrift sequence was interpreted based on core & cutting samples and biostratigraphy study of 4 wells. Data from wells used in this study include gamma ray and resistivity logs from four wells named by the research as WELL 1, WELL 2, WELL 3 and WELL 4.

DEPOSTIONAL ENVIRONEMNTS

Depositional environments are defined in a various ways; however, all definitions of environment have in common an emphasis on the physical, chemical, and biological conditions of the environment. A depositional environment is thus characterised by a particular geomorphic setting in which a particular set of physical, chemical and biological processes operate to generate a certain kind of sedimentary deposit.

➢ *FLUVIAL SYSTEMS*

Fluvial Systems encompass a wide spectrum of sediments generated by the activities of rivers, streams, and associated gravity-flow processes.

ALLUVIAL FANS

Alluvial fans are the deposits with gross shape approximately a segment of a cone and exhibiting convex up cross sectional profile. Sediments are typically poorly sorted, gravel-size detritus (debris/waste), sudden change in gradient in the areas of high relief thus abundant supply of sediment is available. Alluvial fan is of two types- debris-flow dominated fans and stream-flow dominated fans. Also, sediment gravity flows, including debris flows and mud flows, are dominated transport and depositional processes on many fans in both arid-semiarid regions and humid settings.

Debris-flow deposits are poorly sorted, shows reverse graded bedding, lacks sedimentary structures and are non-porous. Landslides are commonly associated with debris flows and in many cases landslide deposits forms a source of sediment for the debris flows. The surface of debris-flow-dominated fans tends to be steep with little vegetation and characterised by lobes of poorly sorted, coarse sediments, commonly associated with a muddy matrix. The stream-flow-dominated fans constitutes more like sheet-like deposits of gravel, sand & silt that may be moderately well sorted and show cross-bedding with laminated or nearly structure-less.

Overall, alluvial-fan deposits tend to be characterised by strongly developed thickening and coarsening upward successions caused by active fan progradation/outbuilding. Nonetheless, some fans display thinning and fining upward succession, which indicates relative inactivity of depositional processes or fan retrogradation/retreat.

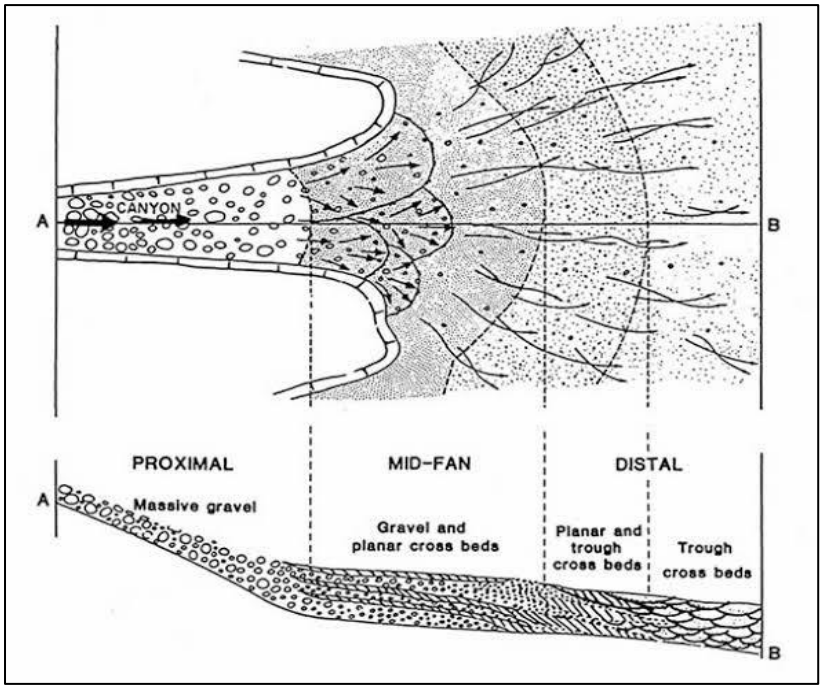


Figure 1: Proximal and Distal division of an alluvial fan (Source: Sam Boggs Jr 1995)

RIVER SYSTEMS

To recognise & understand the deposits of ancient river systems, it is useful to examine the channel shapes, sediment transport processes, and sediment characteristics of a modern river.

***Channel form*:** Anostomosing river- It is composed of two or more interconnected channels that encloses floodbasins, Meandering river- a single channel river (sandy and silty), Braided river- multiple channel river (sandy and gravelly).

***Sediment transport processes in river*:**

Channel transport- Sediment transport (and erosion) in the higher gradient, proximal reaches of rivers occurs mainly within the river channels. Downstream flow of water around channel bends leads to helical spiralling of flow, out towards the surface and inward the bed. The channels are characterised by the present of the *bars*. *Point bars* are attached to the river bank. The helical flow of water erodes on the outside parts of bends, and deposition on the point bars which occurs via lateral accretion. Cross-bedding and general fining upward occurs toward the top of the bar. Braid bars(channel bars/medial bars/longitudinal bars/transverse bars) present in the mid channel positions. The helical flow splits around the upstream end of the bar, helical flow causes lateral accretion on both the sides of the bar,

showing upward fining sequence of deposition. *Braid bars* are free to move , delta like deposition takes place at the downstream end of the bars. Braid bars can migrate downstream.

Floodplain deposition

Natural-Levees deposits – form primarily on the concave/steep side of meander loops immideately adjacent to the channel as a result of sudden "loss of competence". And they typically contain horizontally stratified fine sands overlain by laminated mud.

Crevass-Splay deposits – It occur on floodplains where rising floodwater breach natural levess. Sedimentation from traction and suspension occurs rapidly after breaching as water containing both coarse bedload sediment and suspended sediment debouches suddenly onto the plain, resukting in graded deposits that may resemble a Bouma turbidite sequence.

Characteristics of fluvial depostits: It is clear from the preceding discussion that sediments can be deposited in a variety of subenvironments within the fluvial system: on point bars and channels of meandering rivers, in braid bars of braided rivers, and in natural levees, flood basins, and oxbow lakes of floodplains. Fluvial deposits have some common properties. Most fluvial deposits consist of sand and gravel, although mud maybe common in floodplain deposits. Sorting of most fluvial sediments ranges from moderate to poor. The deposits of point bars and braid bars generally display fining-upward grain size owing to the helical nature of sediment transport on bars. Migration of meanders also produce fining upward sequence. Fluvial deposits shows abundant traction structures, including planar and trough cross bedding, upper flow regime planar bedding, and ripple-marked surfaces.

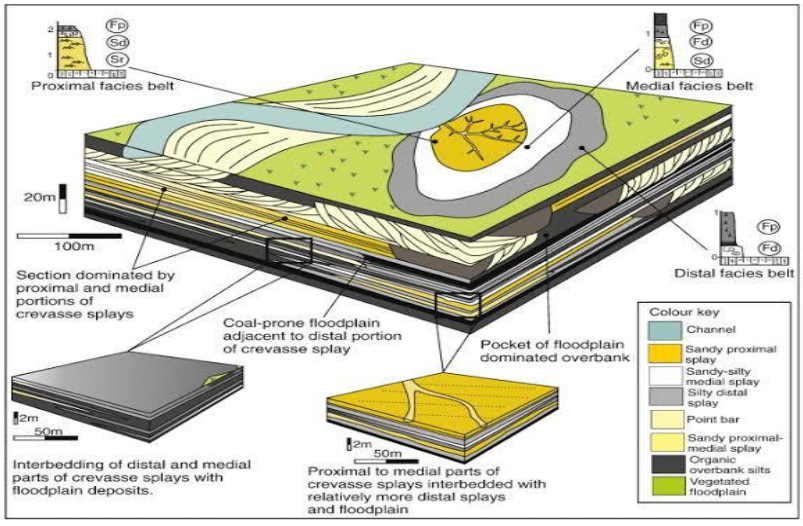


Figure 2: A 3-dimensional model illustrating meandering river system and its associated depositional structures.

- ***DELTAIC SYSTEMS***

Delta is defined as a body of sediments deposited when a river flow into standing water (sea or ocean). They are triangular in view in plain cross section .The shape of delta is influenced by sediment input, wave and tidal energy. Grain size ranges from coarse sand to fine mud, generally becoming finer away from land. Common structures such as small cross-beds, ripple marks and organic material is common in delta. Morphological units of delta are- Delta plain, Delta front and Prodelta.

There are three types of deltas- 1. River dominated delta

2. Wave dominated delta

3. Tide dominated delta

Deltas - form where rivers enter a standing body of water, slow down, and deposit more sediment than can be removed by waves and currents. Deltas are composed of several sub-environments, from the fluvial delta-top to the submarine base of the delta. Accordingly there are numerous types of fossils and sedimentary structures possible. . Because of their formation process, there is a lateral gradation in particle size (and sedimentary rock type) along the delta, from sand near the river outlet to submarine clay deposits at the edges. As more sediment is added, the delta builds out into the standing body of water, with coarser sediments migrating across the clays that used to be at the delta edges. This results in a coarsening upward (regressive) sequence.

The formation of a delta is complicated, multiple, and cross-cutting over time, but in a simple delta three main types of bedding may be distinguished: the bottomset beds, foreset/frontset beds, and topset beds. This three part structure may be seen in small scale by crossbedding.

The bottomset beds are created from the lightest suspended particles that settle farthest away from the active delta front, as the river flow diminishes into the standing body of water and loses energy. This suspended load is deposited by sediment gravity flow, creating a turbidite. These beds are laid down in horizontal layers and consist of the finest grain sizes.

The foreset beds in turn are deposited in inclined layers over the bottomset beds as the active lobe advances. Foreset beds form the greater part of the bulk of a delta, (and also occur on the lee side of sand dunes). The sediment particles within foreset beds consist of larger and more variable sizes, and constitute the bed load that the river moves downstream by rolling and bouncing along the channel bottom. When the bed load reaches the edge of the delta front, it

rolls over the edge, and is deposited in steeply dipping layers over the top of the existing bottomset beds. Under water, the slope of the outermost edge of the delta is created at the angle of repose of these sediments. As the foresets accumulate and advance, subaqueous landslides occur and readjust overall slope stability. The foreset slope, thus created and maintained, extends the delta lobe outward. In cross section, foresets typically lie in angled, parallel bands, and indicate stages and seasonal variations during the creation of the delta.

The topset beds of an advancing delta are deposited in turn over the previously laid foresets, truncating or covering them. Topsets are nearly horizontal layers of smaller-sized sediment deposited on the top of the delta and form an extension of the landward alluvial plain. As the river channels meander laterally across the top of the delta, the river is lengthened and its gradient is reduced, causing the suspended load to settle out in nearly horizontal beds over the delta's top. Topset beds are subdivided into two regions: the upper delta plain and the lower delta plain. The upper delta plain is unaffected by the tide, while the boundary with the lower delta plain is defined by the upper limit of tidal influence.

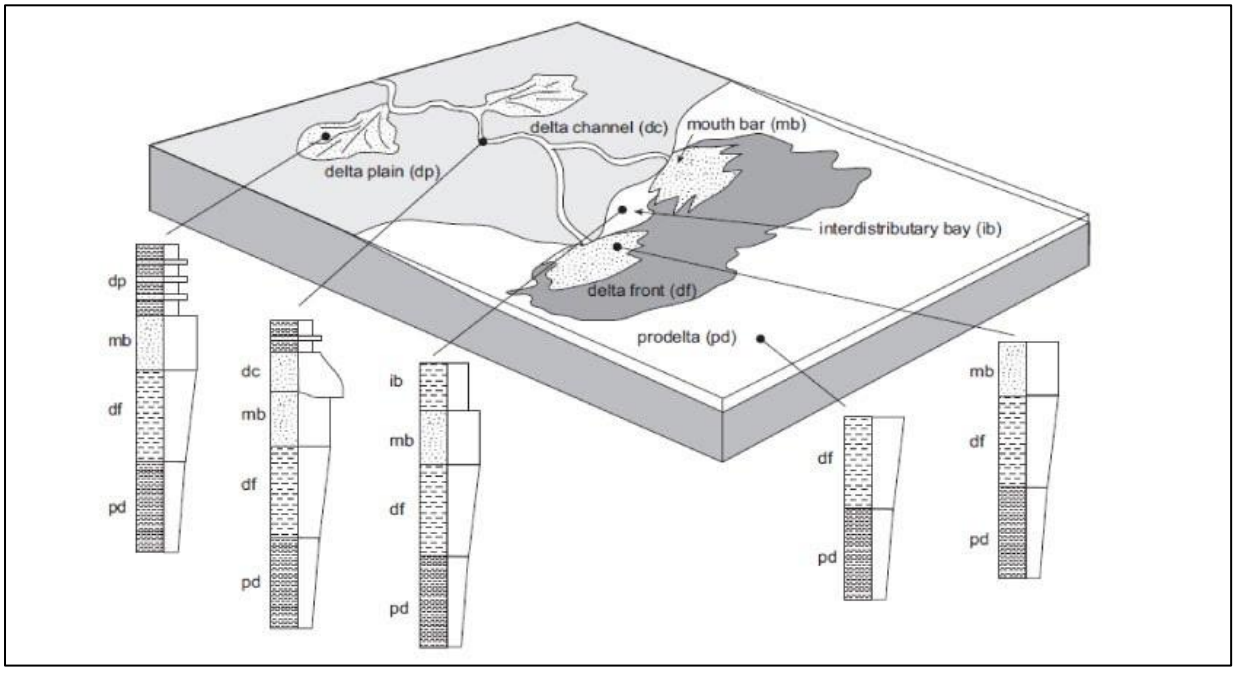


Figure 3: Delta Front(df) and prodelta(pd) depositional sequemce.

➢ ***BEACH & BARRIER ISLAND SYSTEMS***

Mainland beaches are long, narrow accumulation of sand aligned parallel to the shoreline and attached to land. Bodies of beach sand are typically cut across here and there by headlands and sea cliffs, estuaries, river deltas, tidal inlets, bays and lagoons. *Barrier-island beaches are similar to mainland beaches but are separated from land by a shallow lagoon, estuary, or marsh.* Ancient beach & barrier-island sediments have considerable economic importance as reservoirs for petroleum & natural gas.

Barrier-Island Systems is not a single environment but a composite of three separate environments: the sandy barrier-island chain itself (the subtidal to subaerial barrier-beach complex); enclosed lagoon, estuary, or marsh behind(the back barrier, subtidal-intertidal region) &; the channels that cut through the barrier end connect the back-barrier lagoon to the open sea (the subtidal-intertidal delta and inlet-channel complex).

Sediment transport & depostional processes on barrier beaches are similar to those on mainland beaches; however; additional processes take place in tidal channels, tidal flats, marshes, and lagoons of barrier complexes.

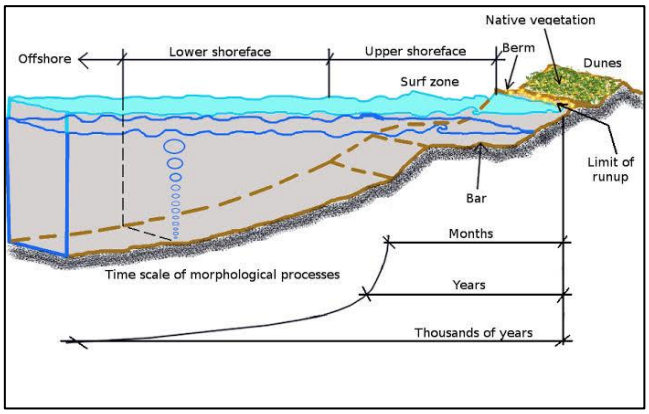


Figure 4: Generalized cross-sectional profile of the beach and nearshore zone, showing also the principal zones of wave activity (Sam Boggs Jr. 1995)

Depostional settings: Beach and Barrier-island complexes are best developed on wave-dominated coastes where tidal range is small to moderate. Repeated deposition and reentrainment of sediment in the beach regime tends to winnow and remove the finest sediment, producing generally well sorted positively skewed deposits. Beach Deposits form on the beach face or foreshore, which is intertidal zone extending from mean low-tide level to mean high-tide level, corresponding to the zone of wave swash. Sediments of the **foreshore** consist predominantly of fine to medium sand but may also include pebbles and gravel lenses

or layeser. Sedimentary structures are mainly parallel laminae, formed during swash-backswash flow, that dip gently (2 to 3 degrees) seaward. Thin, heavy minersal laminae are commonly present, alternating with layers of quartzose sand. Thin lenticular sets of low-angle, lamdward dipping laminae, possibly formed by antidune migration during backwash, maybe present also. Some foreshore sand may display high angle cross-beds. **Backshore** is inundated onkly during storms and is thus a zone dominated by intermittent storm-wave base deposition. Faint, landward-dipping, nearly horizontal laminae, interrupted locally by crustacean burrows, record deposition by storm waves. **Shoreface** deposits extends from mean-low-tide level to mean fair weather wave base the depth of the wave base on the shoreface is commonly on the order of 10m-15m, but this depth can be lowered significantly during storms. The order is divided into upper, middle, and lower shorefaces. *Upper-shoreface* (surf zone) deposits form in an environment dominated by bstrong bidirectional translation waves and longshore currents. Multidirectional trough cross-bed sets. *Middle-shoreface (breaker zone)*- high energy conditions due to breaking waves and associated longshore and rip currents. Sediments- fine to medium grained sand, both landward and seaward dipping trough cross-beds as well as sub-horizontal plane laminations. *Lower-shoreface (outer shoaling zone)* deposits form under relatively low-energy conditions and grade seaward into open-shelf deposits. They are commonly composed dominantly of fine to very fine sand, but may contain thin, intercalated layers of silt and mud. Sedimentary structures can include small-scale cross stratification, planar, nearly horizontal laminated bedding; and hummocky cross-stratification under stormy conditions.

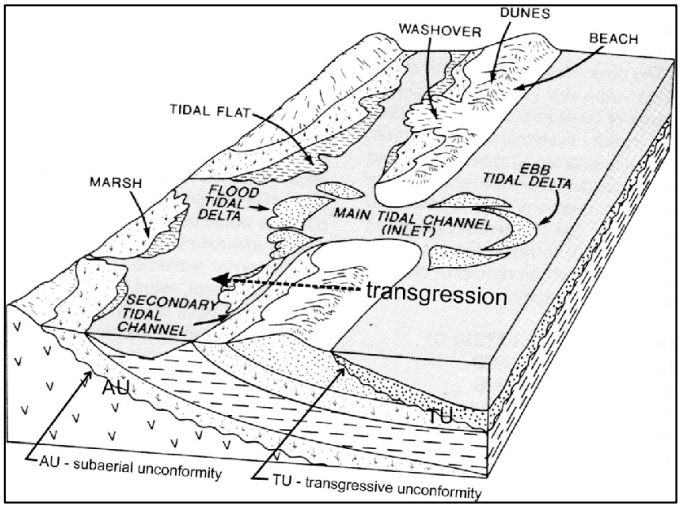


Figure 5: A 3-d model illustrating beach-barrier island environment of deposition (Adapted from Sam Boggs Jr. 1995)

➢ ***ESTUARINE SYSTEMS***

Estuaries are the seaward potion of a drowned valley system which receives sediment from both fluvial and marine sources. It contains facies influenced by tide, wave and fluvial processes. Estuaries can form only in the presence of a relative sea-level rise/transgression. Progradation tends to fill and destroy estuaries causing them to change into deltas. Estuaries are of three type- wave-dominated, tide-dominated and mixed wave-and-tide-dominated. Wave dominated estuaries are characterised by the presence of a barrier bar that extends across the moputh of the estuary and partially or completely blocks the estuary. Tide-dominated estuaries are characterised by the presence of bars that extent longitudinally into the estuary but do not completely block the mouth.

Estuarine Facies- Depending upon the location within the estuary, estuarine deposits may consist almost entirely of cross-bedded sands, laminated or bioturbated muds or combinations of sand and mud. Gradation from fluvial channel sand at the base of a vertical section through mixed fluvial marine muds in the middle of the section to marine (tidal) sands at the top suggests transgressional estuarine deposits. Facies of Estuaries are dominated by cross-bedded, bioturbated sandstone present near the mouths of estuaries and in fluvial-tidal channels; laminated to well-bioturbated muds occupy the non-channel middle and upper part of estuary.

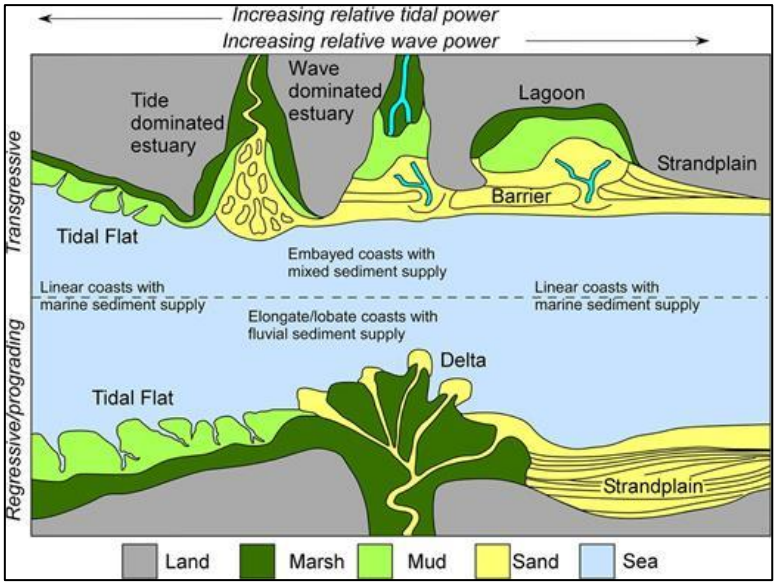


Figure 6: The principal coastal environments of the marginal-marine depositional setting. The figure is organized to show the relative influence of tidal power (increasing to the left) and wave power (increasing to the right) on each environment. Deltas are features of progradaing (regressive) coast, whereas estuaries and lagoons are particularly characteristic of transgressive coast. (Source: Sam Boggs Jr.1995).

➢ ***LAGOONAL SYSTEMS***

A coastal lagoon is defined as a shallow stretch of seawater- such as a sound, channel, bay, or saltwater lake- near or communicating with the sea and partly or completely separated from it by a low, narrow elongate strips of land, such as a reef, barrier island, sand bank, or spit. Most modern lagoons are formed behind spits/offshore barriers of some type and thus are elongated bodies lying parallel to the coast with a narrow connection to open ocean. Many lagoons have no significant freshwater runoff: lagoons may occur in close association with river deltas, barrier island, and tidal flats. Factors affecting water flow, water missing and sediment transport in lagoons:- tides, wind waves, freshwater runoff, episodic storms, density gradients, sea-level changes and changes in climate and temperature. Lgoons are predominantly areas of low water energy. Tidal deltas commonly develop at the ends of these tidal inlets both within the lagoon and on the ocean sides, and sandy sediments may also be deposited within the higher energy tidal channels inside the lagoon. Sedimentation within lagoons is dominated by deposition of silt and mud, although occasionally high wave activity during storms can cause wash over of sediment from the barrier. Sediment deposited in lagoons are derived (depending upon the nature of lagoon) river, the ocean, shores, barriers under typically low energy environment, but storms provide occasional episodes of high-energy waves that wash over barriers into lagoons. Deposition of fine-grained sediments bottom is covered with silty or muddy sediments, commonly extensively bioturbated that may contain thin intercalations of sand brought in by storms or blown in by wind. This sand is horizontally laminated but it may display ripple cross-lamination.

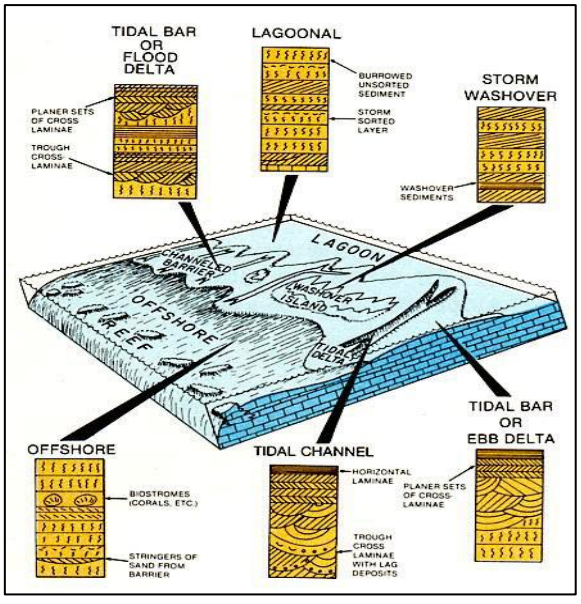


Fig 7: A 3-D model illustrating depositional environment & its associated sed. structures in a lagoonal system (source: M E Tucker)

- ***TIDAL FLAT SYSTEMS***

Tidal flats form primarily on *mesotidal(1-4m), macrotidal(10-15m)* coasts, where strong wave action is absent. They develop either along open coates of low relief and relatively low wave energy or behind barriers on high-energy coasts where protection is afforded from waves by barrier islands, spits, reefs and other structures. Therefore, they occur within estuaries, bays, the backshore of barrier-island complexes, and deltas, as well as along open coasts. Marshy and muddy-to-sandy areas that are alternatively covered and uncovered by the rise & fall of tides. Tidal flat environment is divided into 3 zones – subtidal zone (lies below mean low tide level), intertidal zone (lies between mean high and low tide level), supratidal zone (lies above normal high-tide level but is incised by tidal channels and flooded extreme tides).

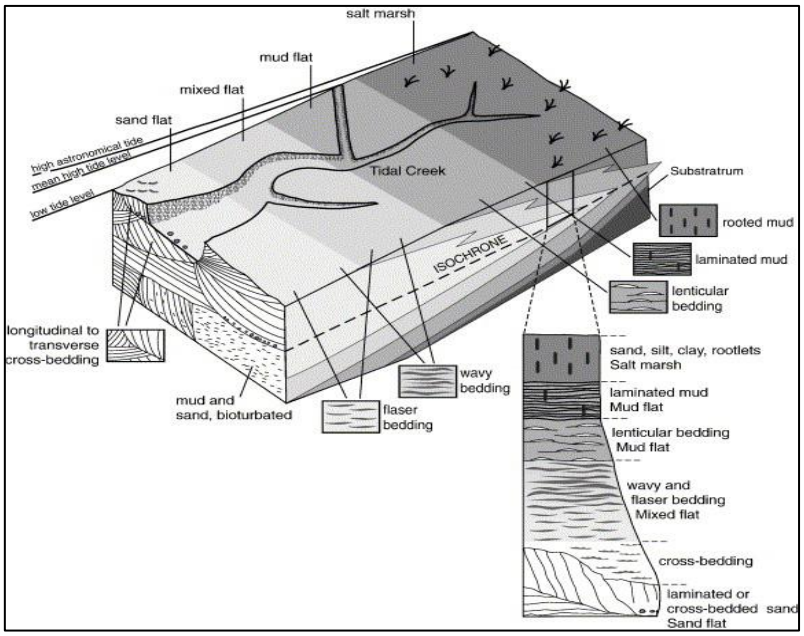


Figure 8: A 3-D model illustrating depositional environment & its associated sed. Structures in a tidal-flat system (Adapted from Sam Boggs, Jr.1995)

Ancient Tidal-Flat Sediments:

The most important criteria for recognition of ancient tidal flat deposits are commonly regarded to include the following:- bimodal direction of current-formed cross-bedding resulting from reserved tidal currents; the occurrence of sedimenrs that reflect repeated, small-scale alterations ion sediment transport conditions, abundant reactivation surfaces and flaser bedding, a high frequency of erosional contacts and abrupt facies changes.

INTERPRETATION OF DEPOSITIONAL ENVIRONMENTS FROM WELL LOGS

Gamma ray is frequently an indicator of clay (shale) content. Gamma ray log shapes can be explained in terms of variations in clay content. The SI unit of measuring Gamma ray log is API (American Petroleum Institute).

Gamma ray increases regularly upwards from a minimum value should indicate *increasing clay content.* Gamma ray ray log and sandstone grain size show close relationship on gamma ray log. Therefore, gamma ray log can be interpreted as grain-size trends and, by sedimentological association, as facies successions. Gamma ray value decreases as grain size increases (coarsening) & vice versa. A bell shape in gamma ray log indicates "alluvial/channel fan" but also a "transgressive shelf sand". And a funnel shape is a coarsening upward succession while may be a "deltaic progradation" or a shallow marine progradation.

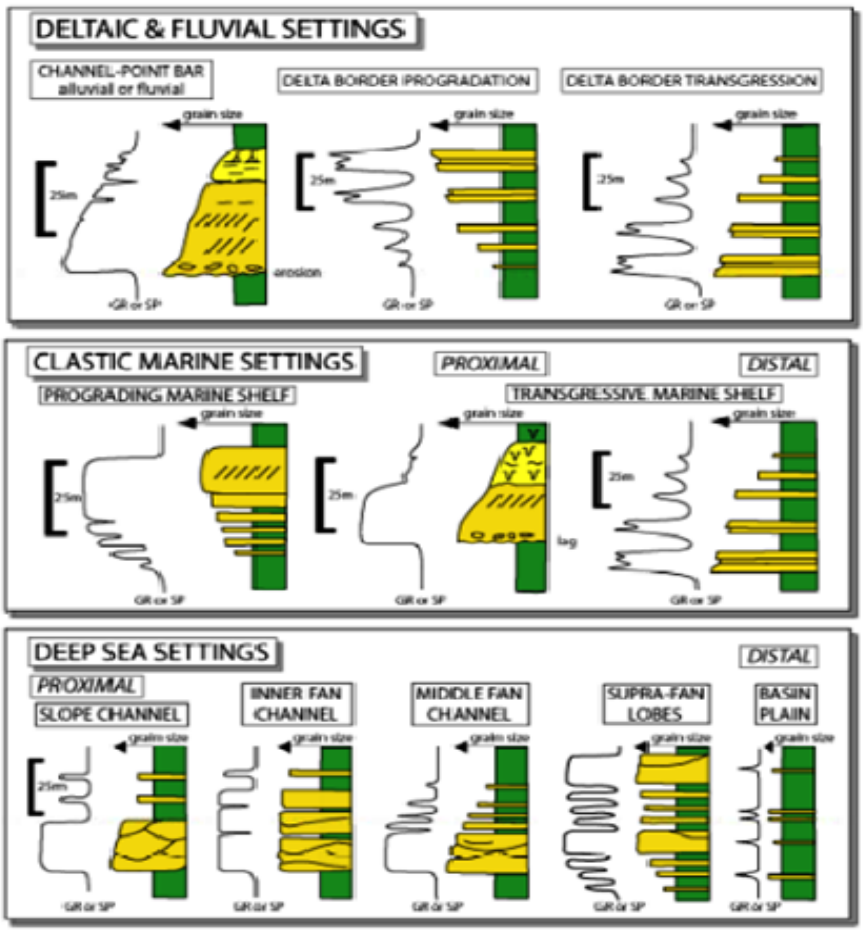


Figure 9: Gamma ray log shapes and depositional settings (Adapted from Rider, MH 1999)

The **resistivity log** is a measurement of a formation resistivity, i.e. resistance to the passage of an electric current. It is measured by resistivity tools. Most rock materials are essentially insulators, while their enclosed fluids are conductors. Hydrocarbons are exception to fluid conductivity, and on the contrary they are infinitely resistive. When a formation is porous and contains salty water then the overall resistivity will be low, and when the formation contain hydrocarbon resistivity will be very high.

The **resistivity logs** were developed to find hydrocarbons and it is also used in petrophysical calculations. Formations resistivity is one of its typical geophysical characteristics and as such can contribute information on lithology, texture, facies, over pressure and source rock aspects. Log is frequently used for correlation.

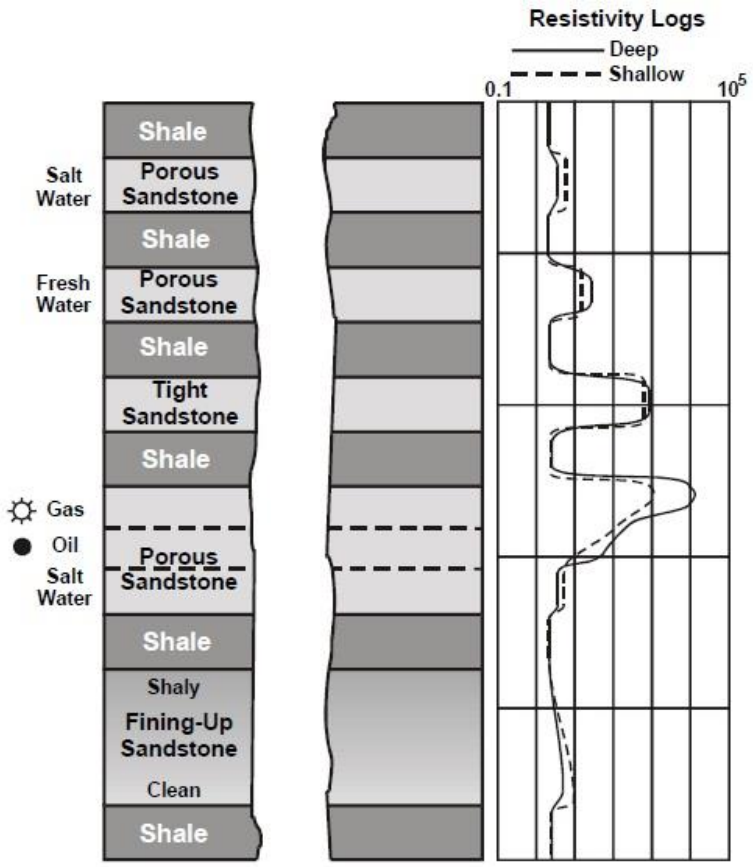


Figure 10: Resistivity logs (Adapted from Rider, MH 1999)

CASE STUDY

This study has been carried out in the synrift section of the offshore field of the KG basin. In this study, the determination of the lithology of the synrift sequence has been done on the basis of gamma ray characteristics. Depositional environment and age of the synrift sequence was interpreted based on core & cutting samples and biostratigraphy study of 4 wells. Data from wells used in this study include gamma ray and resistivity logs from four wells named by the research as WELL 1, WELL 2, WELL 3 and WELL 4.

LOCATION & GEOLOGY OF THE STUDY AREA

Krishan-Godavari basin (KG basin) is a peri-cratonic passive margin basin on the east coast of India (Figure 11). The onland part consists of 28000 km^2 and is mostly alluvium covered. Krishna and Godavari are the two major river systems which drain the area and discharge in the Bay of Bengal. The offshore basinal area covers 24000 km^2 to the isobath of 200 m. However, the basin extends into deeper water and covers a much larger area. The basin's characteristic feature is its en echelon horst and graben system which is filled with a thick pile of sediments of Permian-to-Recent age and emerging as one of India's most promising petroliferous areas. Commercial accumulation of hydrocarbons occurs in sediments from the Permian to as young as the Pliocene.

KG basin is orthogonally juxtaposed to NW-SE trending Pranhita Godavari Gondwana graben in the north (Figure 11). The NE-SW basin margin is the most extensive fault trend over the area. It takes a bow-like swing and comes to the coast near Kakinada graben in the northeast and near Palar-Pennar graben in the southwest. The onland basinal area is characterized by en-echelon and arcuate horsts and grabens associated with major cross trends. In addition to the basin margin fault, three more regional faults developed further basinward: the onland Matsyapuri-Palakollu fault, a Miocene structure building fault in shallow water close to the coast, and a Pliocene structure building fault in deeper water. The arcuate horsts and the four regional arcuate faults are more or less parallel. Offshore, the sediments are mostly influenced by growth related tectonic.

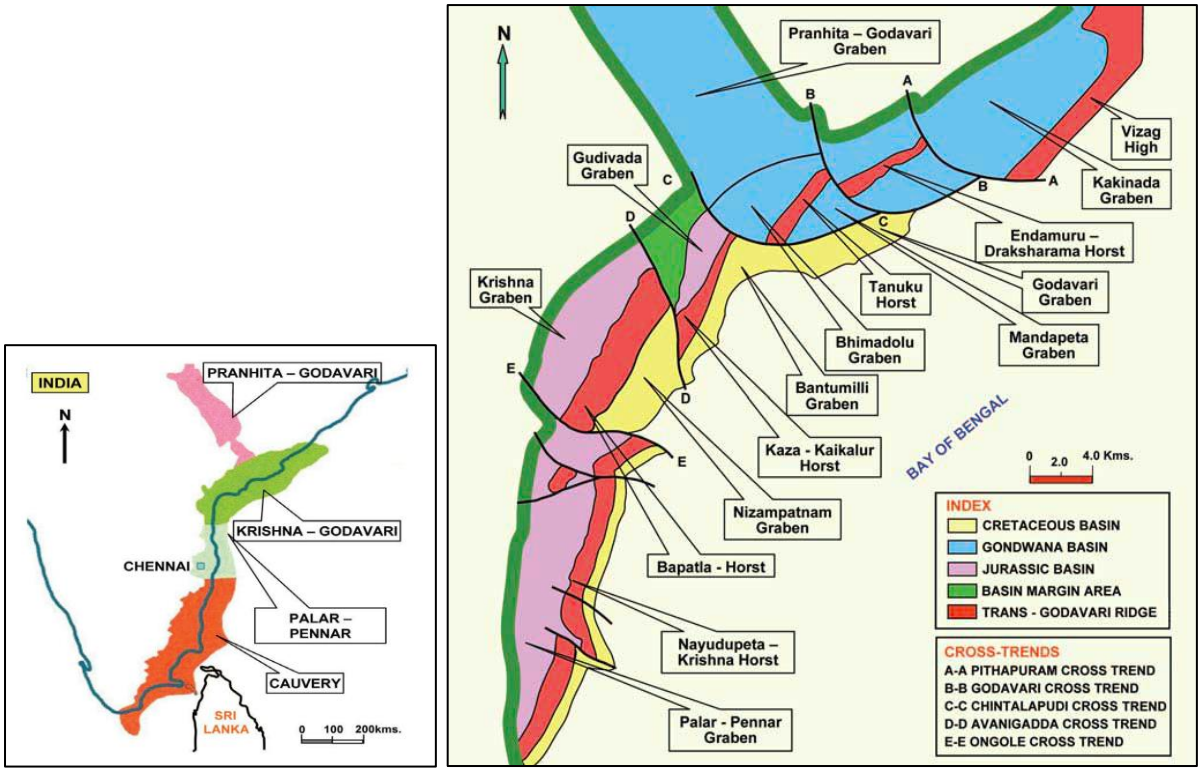


Figure 11: Geologic location & tectonic map of Krishna Godavari Basin (After S.K Gupta, 2006)

STRATIGRAPHY OF THE STUDY AREA

The sedimentary sequence of Krishna Godavari Basin ranges from Permian-to-Recent. The Precambrian metamorphic basement consists of gneisses, quartzites, charnokite, and khondalite. Subsurface knowledge is limited due to alluvium cover. However, isolated outcrops of Permian, Cretaceous, Paleocene, and Mio-Pliocene rocks are present near the basin edge. Figure 12 shows the geologic map of Krishna Godavari Basin and Figure 13 the generalized stratigraphy. Sedimentation in Gondwana Basin was initiated during the Early Permian over the crystalline basement and is known as Draksharama/Kommugudem Formation. Sediments of Triassic age are conspicuous by their absence. Golapalli sandstone of Early Cretaceous age lies unconformably over Mandapeta sandstone.

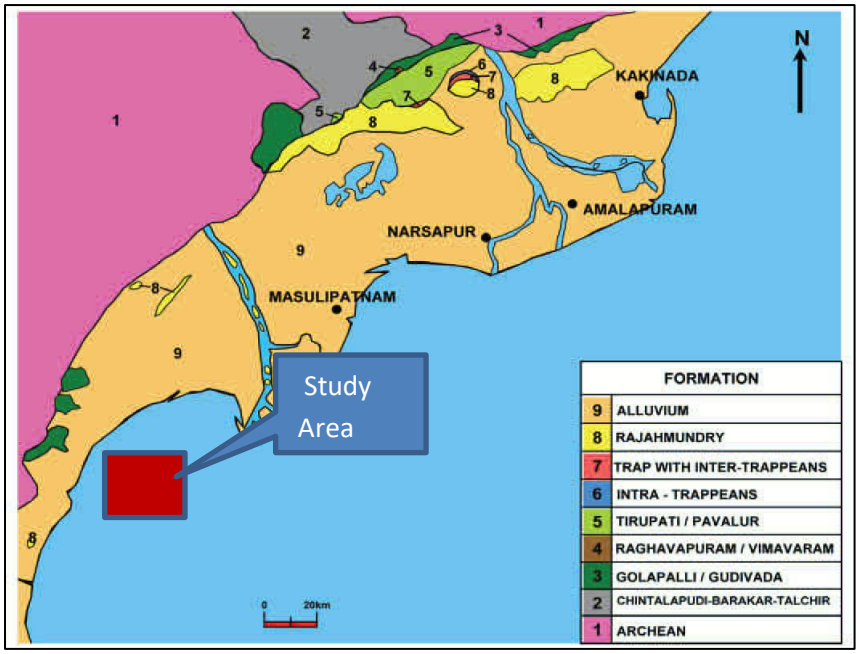


Figure 12: Geologic Map of Krishna Godavari Basin (After S.K. Gupta, 2006)

The top of the Golapalli sandstone is a basinwide regional unconformity and is in turn covered by thick Late Cretaceous sediments and relatively thin Tertiary sediments. The areas of Gudivada graben and Krishna graben constitute the Jurassic rift basin. The Jurassic and Early Cretaceous sediments are the main fill for the Jurassic Basin. The overlying Tertiary and Recent sediments are relatively thin and generally undifferentiated. Bantumilli graben and Nizampatnam graben were created during the Cretaceous and therefore named the Cretaceous Basin. The Cretaceous sedimentation over the Gondwana, Jurassic, and Cretaceous basins began simultaneously during Aptian/Albian time. These sediments are called the Golapalli sandstone in Mandapeta graben and Bhimadolu graben, the Nandigama Formation in Bantumilli graben, and Gajulapadu shale/Kanukollu sandstone in Gudivada graben and Krishna graben. The overlying Raghavapuram shale and Tirupati sandstone of Late Cretaceous age, followed by the relatively thin Tertiary sediment is widely distributed. Therefore it can be surmised that Krishna Godavari Basin was initially made up of Gondwana Basin, Jurassic Basin, and Cretaceous Basin, and they remained a unified basin until the Late Cretaceous. The Godavari graben area to the south of the Eocene growth fault (Matsyapuri-Palakollu fault) is the main depocenter for Tertiary sediments. Further basinward, Tertiary sedimentation was successively influenced by the Miocene growth fault and the Pliocene growth fault. Each kicked off a series of subsidiary growth faults which locally guided the sedimentation pattern.

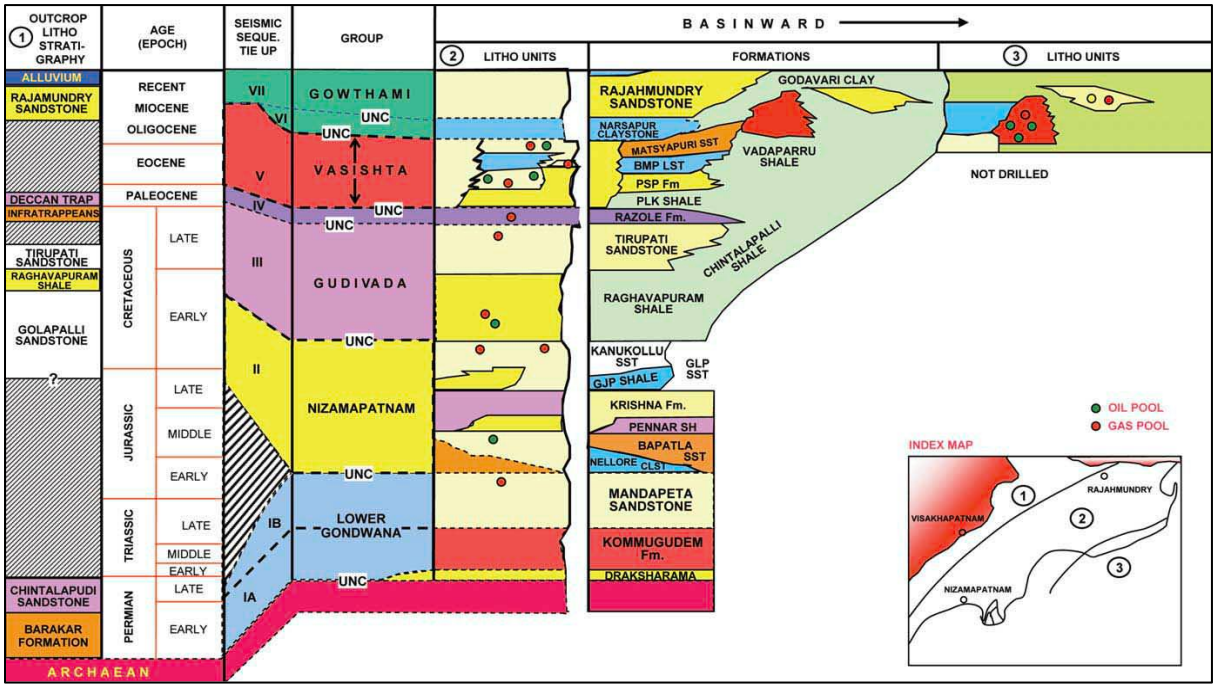


Figure 13: Generalized Stratigraphy (S.K.Gupta ;ONGC, Dehradun Basin Architecture & petroleum system of KG Basin)

INTERPRETATION OF DEPOSTIONAL ENVIRONMENT IN THE STUDY AREA

The study has been carried out in the synrift section of the offshore field of the KG basin. In this study, the determination of the lithology of the synrift sequence has been done on the basis of gamma ray characteristics. Depositional environment and age of the synrift sequence was interpreted based on core & cutting samples and biostratigraphy study of 4 wells. Data from wells used in this study include gamma ray and resistivity logs from four wells named by the research as WELL 1, WELL 2, WELL 3 and WELL 4.

The synrift section in the wells have been divided into 3 units, i.e., unit –III, unit-II, and unit-I on the basis of difference in gamma ray characteristics which would differ depositional settings in all the three units.

The description of the depositional settings and lithological and mineralogical variation in all the units has been detailed below.

WELL-1

UNIT-III

Gamma ray log of Unit-3 Well-1 covers the depth of 88m from 3727m-3815m shows box type of sandstone with intercalations of claystone. The sedimentological report suggests that this unit consist of multistack sandstone bodies, texturally, mineralogically immature, calcareous and calcareous and organic matter rich. Sandstone deposits are of rift fill nature having extreme grain size, ill-sorted being deposited proximal to source area as **braided fluvial bars** with paralic imprints.

High degree of compaction, extreme hardness, ill-sorted nature has resulted in tight reservoirs facies. Mostly porosities are isolated in nature.

UNIT-II

Unit 2 of well-1 covers 3815m to 3985m. It consists of relatively fine grained, moderately sorted Micro Facies: Arkosic wacke. Presence of organic matter/asphalt in inter granular pore space would impair permeability; alternatively it would help in source to reservoir co-relation.

These reservoir sandstones, underwent excessive diagenetic alteration (late diagenetic stage) evidenced by dominant pore bridging illite, pore lining chlorite along with deformed ductile muscovite/biotite and organic matter a concern for fluid mobility.

Biostratigraphic report suggests it to be deposited in fluvial environment in **early Jurassic** to **late Cretaceous.**

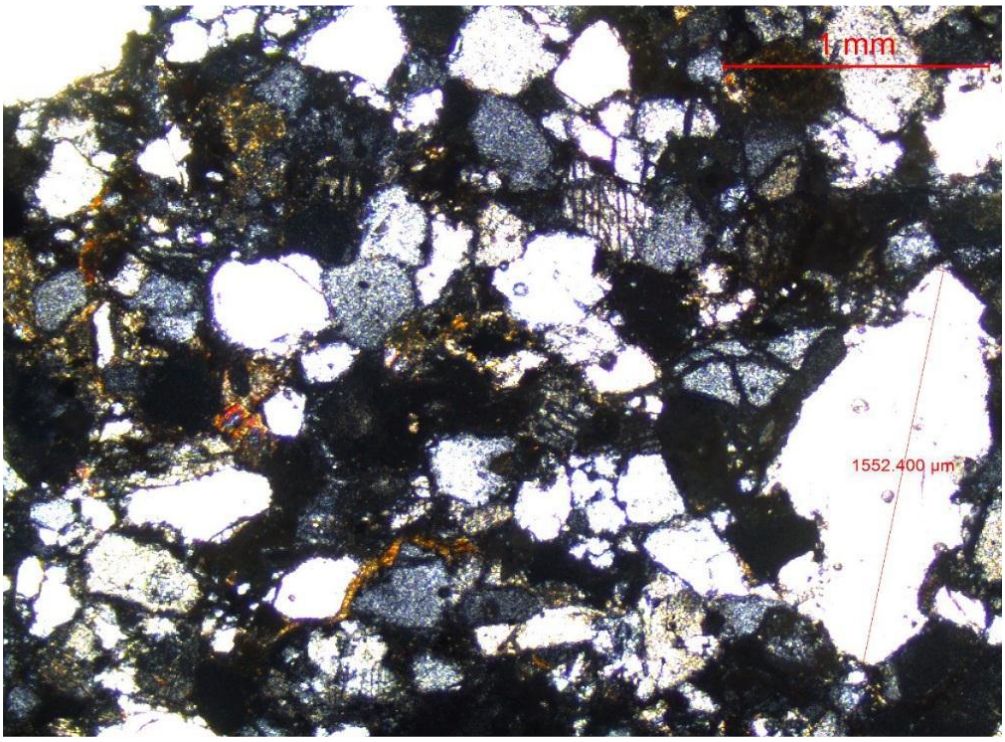


Figure 14: Photomicrographs depicting fine to coarse, occ. pebbly, angular-subangular, poorly sorted, k-feldspar and OM rich Arkosic wacke in WELL-1

WELL-2

UNIT-III

Gamma ray log of unit-III dominantly shows box type sandstone with thin intercalations of shale in between from the depth of 3628-3818m. Sandstone is also showing fining upwards sequence towards the top.

According to the sedimentological report it suggest that **sandstone** is off-white, hard, compact, coarse to pebbly, poorly sorted, feldspar rich matrix, occ. feebly calcareous.

Claystone clasts and asphalt are impregnated at places. Visual porosity is moderate to poor. Though cored portion smells pungent, fluorescence is very faint and not remarkable. Two cycles of deposition:-i) Fining up channel sandstone ii) Amalgamated sandstone.

Shale is dark grey, non-calcareous, micaceous and carbonaceous in nature.

Key sedimentary features include planar cross bedding, sub parallel bedding, abrupt bedding contacts at places where micro faulting is evident and poor sorting collectively indicate **fluvial briaded channel sand.**

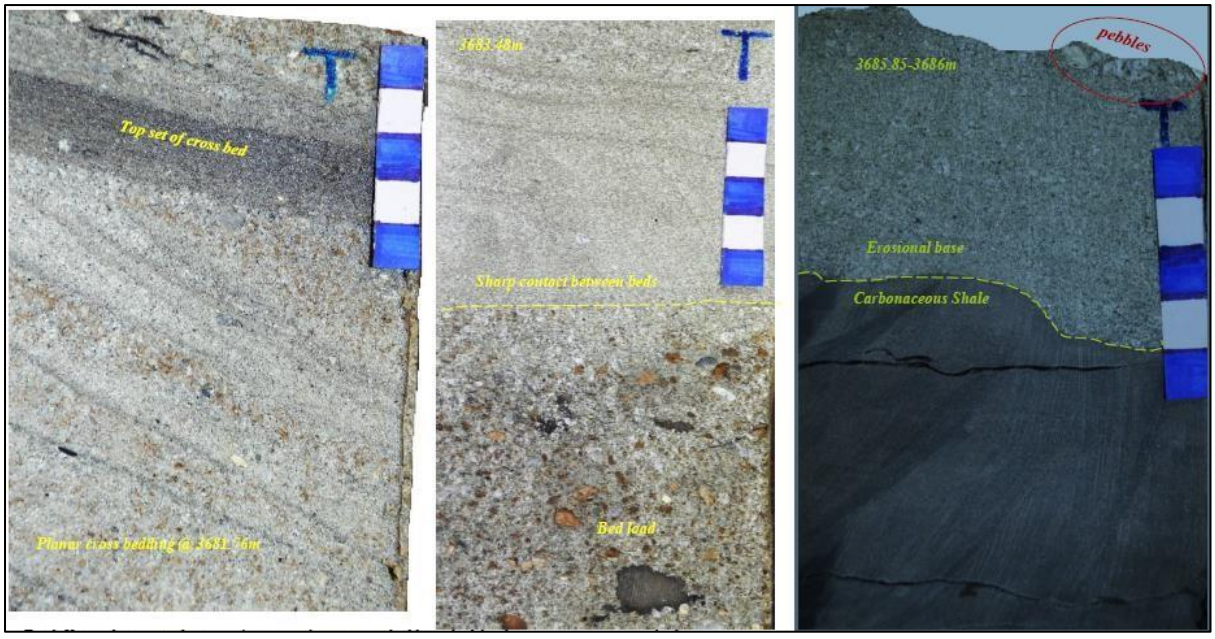


Figure 15: Key sedimentary features showing planar cross bedding; bed load, traction current result in sharp being contacts due to change in grain size, ending with carbonaceous shale, followed by next cycle of sandstone fining up nature. Essentially channel lag with carbonaceous rich overbank facies.

UNIT-II

Gamma log from 3818-4023m shows massive, fining upward thick bedded sandstone with sporadic shale layers.

Sedimentological report shows that sandstone is off white, light grey, fine to pebbly, sub rounded to angular, poorly sorted and feebly calcareous at places; no fluorescence was observed. Visual porosity is moderate.

Extreme grain size is indicative of **braided fluvial-channel** deposit.

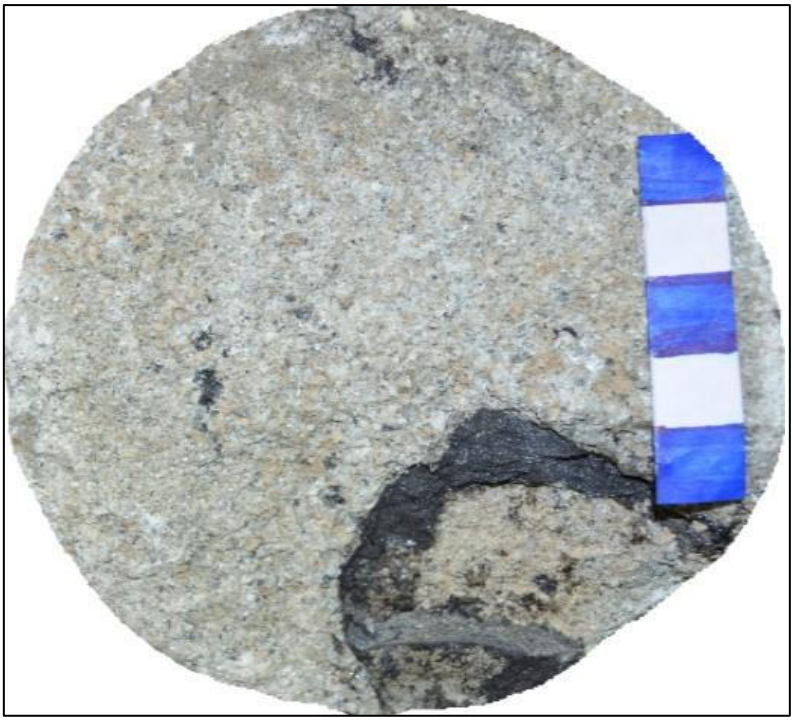


Figure 16: Transverse sections showing relics of organic rich clasts and jet black asphalt rich layers along weak planes(C&D). Sorting is very poor and matrix dominant.

UNIT-I

Gamma log from 4023-4123m shows serrated nature showing thin streaks of sandstone and shale. Core data shows sedimentary features which are flaser bedding, shale clasts, bitumen rich streaks which is suggestive of fluvial channel (lag)-overbank paleo-environments. Petrographic evaluation showed extreme grain size variation, angular to sub rounded, poor sorting, rich in matrix and texturally immature "wacke type reservoir facies".

According to palynological report, **Late Jurassic** to **Early Cretaceous** age has been given to unit –I.



Figure 17: Massive, thick bedded, ill sorted multi stack sandstone (whole core).

WELL-3

UNIT-III

Gamma log from 2838-3216m shows Massive, Box type of sandstone with sporadic shale layers.

Sedimentological report shows that dominant lithology is **massive sandstone** which is off white, light grey, extreme grain variation from fine to pebbly, sub rounded to sub angular, poorly sorted & occ. calcareous in nature. Visual porosity is moderate with thin shale laminations, clasts & lenses are embedded in sandstone with occ. presence of pyrite. Prominent sedimentary features include inclined graded bedding, laminations & micro-faults. Gamma log and sedimentological report shows depositional environment to be **braided fluvial channel type.**

Based on palynological studies, its age is inferred to be **Early Cretaceous**.

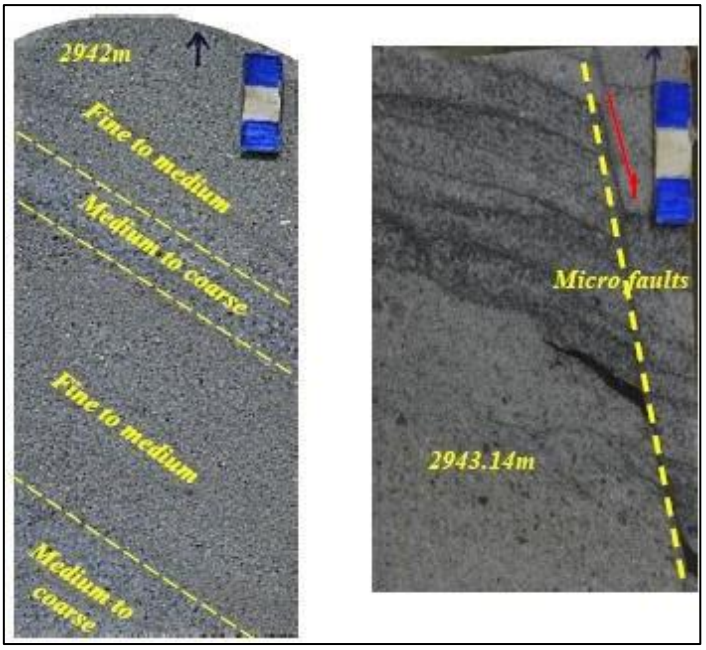


Figure 18: Microfaults



Figure 19: Whole Core, CC-1 Interval:2942-2951m

UNIT-II

Gamma log from 3216-3436m shows major lithology as sandstone showing fining up sequence. sedimentological report shows that dominant lithology is Sandstone which is light grey, fine to medium grained, occ. coarse, sub rounded to sub angular, poorly sorted, non-calcareous, at places rich in mica. Silty shale layers & fractures filled with secondary mineralization at top. Shale clasts are embedded in sandstone. Prominent sedimentary features are laminations & fracture filling.

Gamma log and sedimentological report shows depositional environment to be **braided fluvial channel type.**

Based on palynological studies, its age is inferred to be Early Cretaceous.

UNIT-I

Gamma log from 3436-3686m shows serrated nature showing thin streaks of sandstone and shale. Core data shows Sedimentary features are flaser bedding, shale clasts, bitumen rich streaks suggestive of **fluvial channel (lag) - overbank** paleoenvironments. Petrographic evaluation showed extreme grain size variation, angular to sub rounded, poor sorting, matrix rich sediments. Rock type is suggestive to be texturally immature "wacke type reservoir facies"

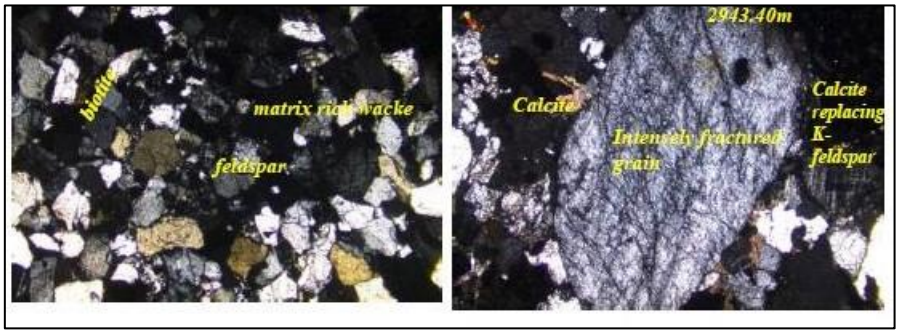


Figure 21: Photomicrographs showing extreme grain size variation, ill-sorted highly fracture, matrix rich, Arkosic wacke reservoir facies. Magnification 2.5x, crossed nicole.

WELL-4

UNIT-III

Unit-III of well 4 covers 3436m-3680m of interval. Gamma log character of upper part of the unit shows alternating coarsening upward and fining upward of sequence which indicates alternate transgressions & regressions, however part of the unit shows box type sandstone which indicates high energy fluvial conditions.

Sedimentological report suggests that this unit consist of multistack sandstone & claystone which is deposited in **fluvial to marginal marine conditions**.

Biostratigraphic report suggests that this unit is dominated by *Pteridophylictrilete* spore and record of fine dinoflagellate cysts which suggest **late Jurassic to early Cretaceous age.**

UNIT-II

Unit-II of well-4 covers 4004m-4224m of interval which shows alterations of fining upward (bell shape) and coarsening upward (funnel shape) sequence. The fining upward sequence suggests either the basin sinking or transgressive marine shelf due to rise in the sea level where as the coarsening upward sequence suggests the **progradation of marine shelf** as a result of falling sea level or regression.

According to the sedimentological report this unit is essentially sandstone with intercalations of claystone & occasional presence of dark coloured organic matter. Sandstone is off-white, fine to coarse grained, non-calcareous, poorly sorted, quartz rich & feldspar (occasionally altered). Claystone is off-white, light grey, soft & non-calcareous.

According to the palynological report, this unit is of **late Jurassic** to **early Cretaceous age**.

UNIT-I

Unit-I of well-4 covers the depth of 200m from 4224m-4424m interval. Gamma ray log represents a sedimentary sequence comprising multiple alternate cycles of beds sandstone & argillaceous beds. This unit shows serrated gamma ray which is dominantly composed of claystone with sporadic sandstone. Sandstone is off-white, fine to medium grained, non-calcareous, poorly sorted, claystone is light grey, off-white, reddish brown, relatively soft, occasional calcareous in nature.

This unit is suggestive to be deposited in **fluvial to marginal marine environment of deposition.**

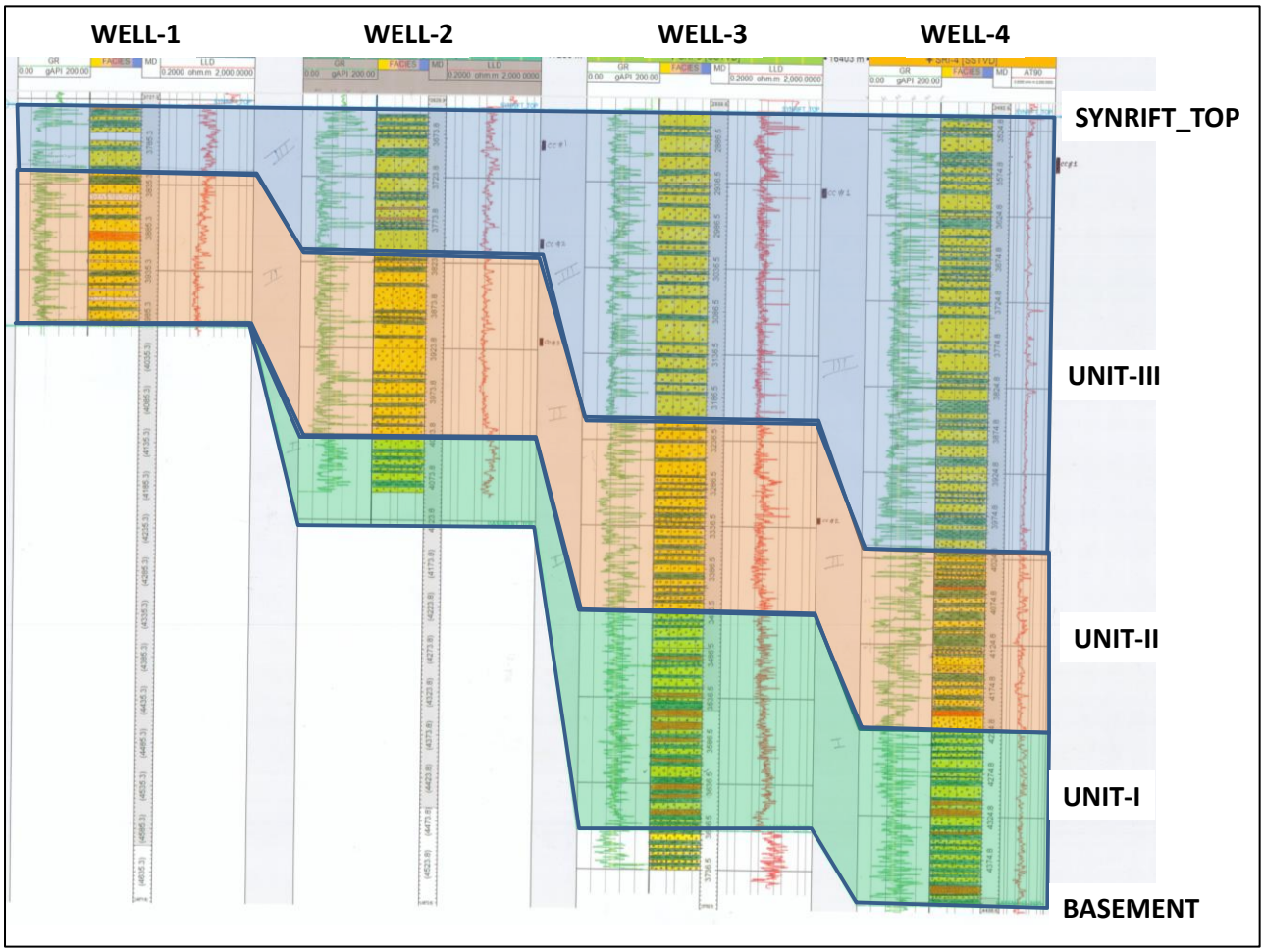


Figure22: Interpretation of lithology and depositional environment from well logs in well-1,2,3 & 4

CONCLUSION

During this summer internship, the depositional environment and lithology of the synrift section has been interpreted on the basis of well log characteristics along with the help of core and cutting description of the sedimentological, biostratigraphic and palynological report.

Gamma ray and resistivity log characteristics play an important role in determining lithology and depositional settings. In the study area of the KG basin, Synrift mainly consist of alternations of sandstone and shale. Gamma ray signatures of the Synrift section mainly show two characterictic attitude, i.e., either fining upward sequence which shows fluvial environment deposition and alternation of alterations of fining upward (bell shape) and coarsening upward (funnel shape) sequence. The fining upward sequence suggests transgressive marine shelf due to rise in the sea level where as the coarsening upward sequence suggests the progradation of marine shelf as a result of falling sea level or regression which suggest marginal marine environment of deposition.

Sedimentological report concludes that Synrift section mainly consist of multiple alternate cycles of beds sandstone and argillaceous beds. Sandstone is light grey, fine to medium grained, occ. coarse, sub rounded to sub angular, poorly sorted, non-calcareous, at places rich in mica. Silty shale layers & fractures filled with secondary mineralization at top. Shale clasts are embedded in sandstone. Claystone is off-white, light grey, soft & non-calcareous. Key sedimentary features include planar cross bedding, sub parallel bedding, abrupt bedding contacts at places where micro faulting is evident and poor sorting. This suggest that synrift is deposited **fluvial to marginal marine environment of deposition.**

According to the palynological studies, Synrift sequence is of **Late Jurassic** to **Early Cretaceous** age.

References:

- *GUPTA, S.K.* (2006) Basin Architecture and petroleum system of Krishna-Godavari Basin, east coast of India.
- *SAHU, J.N.* (2010) Hydrocarbon Exploration Opportunities in Krishna Godavari basin, India
- *P.L. ZUTSHI, M.S. PANWAR* (1997) KDM Institute of Petroleum Exploration Oil & Natural Gas Corporation Ltd. Dehradun; Geology of Petroliferous Basins of India.
- *SAM BOGGS, Jr* (1995) Principles of Sedimentology and Stratigraphy- University of Oregon.
- *TUCKER, M.E.* (2009) Sedimentary Petrology- Depositional Environment
- *B. BIJU-DUVAL* (2002) Sedimentary Geology-Sedimentary Basin Depositional Environment Petroleum Formation
- *RIDER, M.H.* (1999) Geologic Interpretation of Well Logs; Whittles Publishing Services.